# 269--Lab - Insertar, actualizar y eliminar datos en una base de datos

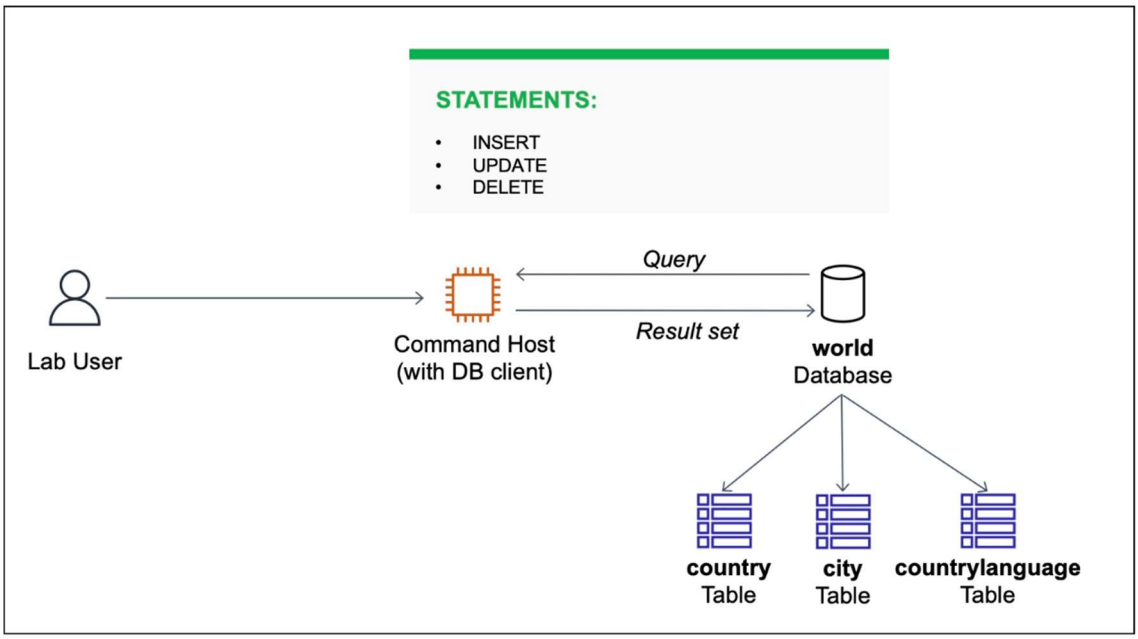
En esta práctica se muestra el funcionamiento de los siguientes comandos en SQL:

- INSERT para insertar registros en una tabla
- UPDATE para modificar datos en una tabla
- DELETE para eliminar registros en una tabla

Asimismo se muestra como ejecutar un script de sql para ejecutar acciones en la base de datos.

## Escenario

Se cuenta con una instancia EC2 configurada con un cliente de base de datos MySQL.




Para conectarnos a la instancia y a la base de datos seguimos los pasos de la práctica anterior.

## Insertar datos en una tabla

Una vez en la terminal de la base de datos mostramos las bases de datos disponibles.

```
MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| world              |
+--------------------+
4 rows in set (0.003 sec)
```

Verificamos que la tabla *country* no tiene registros

```
MariaDB [(none)]> SELECT * FROM world.country;
Empty set (0.001 sec)
```

Verificamos las columnas de la tabla *country*

```
MariaDB [(none)]> DESCRIBE world.country;
+----------------+-----------------------------------------------------------------------------------------+------+-----+---------+-------+
| Field          | Type                                                                                    | Null | Key | Default | Extra |
+----------------+-----------------------------------------------------------------------------------------+------+-----+---------+-------+
| Code           | char(3)                                                                                 | NO   | PRI |         |       |
| Name           | char(52)                                                                                | NO   |     |         |       |
| Continent      | enum('Asia','Europe','North America','Africa','Oceania','Antarctica','South America')   | NO   |     | Asia    |       |
| Region         | char(26)                                                                                | NO   |     |         |       |
| SurfaceArea    | decimal(10,2)                                                                           | NO   |     | 0.00    |       |
| IndepYear      | smallint(6)                                                                             | YES  |     | NULL    |       |
| Population     | int(11)                                                                                 | NO   |     | 0       |       |
| LifeExpectancy | decimal(3,1)                                                                            | YES  |     | NULL    |       |
| GNP            | decimal(10,2)                                                                           | YES  |     | NULL    |       |
| GNPOld         | decimal(10,2)                                                                           | YES  |     | NULL    |       |
| LocalName      | char(45)                                                                                | NO   |     |         |       |
| GovernmentForm | char(45)                                                                                | NO   |     |         |       |
| Capital        | int(11)                                                                                 | YES  |     | NULL    |       |
| Code2          | char(2)                                                                                 | NO   |     |         |       |
+----------------+-----------------------------------------------------------------------------------------+------+-----+---------+-------+
14 rows in set (0.001 sec)
```

Insertamos 2 registros en la tabla *country*

```
MariaDB [(none)]> INSERT INTO world.country VALUES ('IRL','Ireland','Europe','British Islands',70273.00,1921,3775100,76.8,75921.00,73132.00,'Ireland/ire','Republic',1447,'IE
');
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]>
MariaDB [(none)]> INSERT INTO world.country VALUES ('AUS','Australia','Oceania','Australia and New Zealand',7741220.00,1901,18886000,79.8,351182.00,392911.00,'Australia','Co
nstitutional Monarchy, Federation',135,'AU');
Query OK, 1 row affected (0.001 sec)
```

Verificamos los registros en la tabla *country* donde la columna *Code* sea "IRL" o "AUS"

```
MariaDB [(none)]> SELECT * FROM world.country WHERE Code IN ('IRL', 'AUS');
+------+----------+-----------+--------------------------+-------------+-----------+------------+----------------+-----------+-----------+-------------+------------------
-----------------+---------+-------+
| Code | Name     | Continent | Region                   | SurfaceArea | IndepYear | Population | LifeExpectancy | GNP       | GNPOld    | LocalName   | GovernmentForm
                  | Capital | Code2 |
+------+----------+-----------+--------------------------+-------------+-----------+------------+----------------+-----------+-----------+-------------+------------------
-----------------+---------+-------+
| AUS  | Australia | Oceania   | Australia and New Zealand |  7741220.00 |      1901 |   18886000 |           79.8 | 351182.00 | 392911.00 | Australia   | Constitutional Mon
archy, Federation |     135 | AU    |
| IRL  | Ireland   | Europe    | British Islands          |    70273.00 |      1921 |    3775100 |           76.8 |  75921.00 |  73132.00 | Ireland/ire | Republic
                  |    1447 | IE    |
+------+----------+-----------+--------------------------+-------------+-----------+------------+----------------+-----------+-----------+-------------+------------------
-----------------+---------+-------+
2 rows in set (0.000 sec)
```

## Actualizar datos en una tabla

Actualizamos la columna *Population* en la tabla *country* para todos los registros

```
MariaDB [(none)]> UPDATE world.country SET Population = 0;
Query OK, 2 rows affected (0.001 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

Verificamos los cambios

```
MariaDB [(none)]> SELECT * FROM world.country;
+------+-----------+-----------+---------------------------+-------------+-----------+------------+-
-------------------------------------+---------+-------+
| Code | Name      | Continent | Region                    | SurfaceArea | IndepYear | Population |
 GovernmentForm                      | Capital | Code2 |
+------+-----------+-----------+---------------------------+-------------+-----------+------------+-
-------------------------------------+---------+-------+
| AUS  | Australia | Oceania   | Australia and New Zealand |  7741220.00 |      1901 |          0 |
 Constitutional Monarchy, Federation |     135 | AU    |
| IRL  | Ireland   | Europe    | British Islands           |    70273.00 |      1921 |          0 |
 Republic                            |    1447 | IE    |
+------+-----------+-----------+---------------------------+-------------+-----------+------------+-
-------------------------------------+---------+-------+
2 rows in set (0.000 sec)
```

Actualizamos las columnas *Population y SurfaceArea* en la tabla *country* para todos los registros y verificamos los cambios

```
MariaDB [(none)]> UPDATE world.country SET Population = 100, SurfaceArea = 100;
Query OK, 2 rows affected (0.001 sec)
Rows matched: 2  Changed: 2  Warnings: 0

MariaDB [(none)]> SELECT * FROM world.country;
+------+-----------+-----------+---------------------------+-------------+-----------+------------+-
-------------------------------------+---------+-------+
| Code | Name      | Continent | Region                    | SurfaceArea | IndepYear | Population |
 GovernmentForm                      | Capital | Code2 |
+------+-----------+-----------+---------------------------+-------------+-----------+------------+-
-------------------------------------+---------+-------+
| AUS  | Australia | Oceania   | Australia and New Zealand |      100.00 |      1901 |        100 |
 Constitutional Monarchy, Federation |     135 | AU    |
| IRL  | Ireland   | Europe    | British Islands           |      100.00 |      1921 |        100 |
 Republic                            |    1447 | IE    |
+------+-----------+-----------+---------------------------+-------------+-----------+------------+-
-------------------------------------+---------+-------+
2 rows in set (0.000 sec)
```

## Eliminar datos en una tabla

Eliminamos todos los registros de la tabla *country*

```
MariaDB [(none)]> SET FOREIGN_KEY_CHECKS = 0;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> DELETE FROM world.country;
Query OK, 2 rows affected (0.001 sec)
```

El comando *SET FOREIGN_KEY_CHECKS = 0;* sirve para desactivar temporalmente la verificación de claves foráneas en la base de datos. De esta forma el motor de la base de datos permite realizar cambios que normalmente estarían prohibidos cuando se cuenta con columnas que están ligadas a columnas de otras tablas.

Verificamos los cambios.

```
MariaDB [(none)]> SELECT * FROM world.country;
Empty set (0.000 sec)
```

Ahora salimos de la terminal de la base de datos con *QUIT;*

```
MariaDB [(none)]> QUIT;
Bye
```

## Importar datos usando un script SQL

Desde la terminal de la instancia verificamos el contenido del archivo world.sql

```
[root@ip-10-1-11-156 ec2-user]# cat world.sql
-- MariaDB dump 10.19  Distrib 10.6.5-MariaDB, for Linux (x86_64)
--
-- Host: localhost    Database: world
-- ------------------------------------------------------
-- Server version       10.6.5-MariaDB

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;

DROP DATABASE IF EXISTS `world`;
CREATE DATABASE `world` DEFAULT CHARACTER SET utf8mb4;

USE `world`;

--
-- Table structure for table `city`
--

DROP TABLE IF EXISTS `city`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `city` (
  `ID` int(11) NOT NULL AUTO_INCREMENT,
  `Name` char(35) NOT NULL DEFAULT '',
```

El archivo world.sql es un script de SQL que contiene instrucciones para agregar dos tablas adicionales e inserta datos en las tres tablas de la base de datos.

Ejecutamos el script con mysql

```
[root@ip-10-1-11-156 ec2-user]# mysql -u root --password='re:St@rt!9' < /home/ec2-user/world.sql
```

Ahora accedemos nuevamente a la terminal de la base de datos

```
[root@ip-10-1-11-156 ec2-user]# mysql -u root --password='re:St@rt!9'
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 16
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Verificamos las tablas creadas en la base de datos *world*

```
MariaDB [(none)]> USE world;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [world]> SHOW TABLES;
+-----------------+
| Tables_in_world |
+-----------------+
| city            |
| country         |
| countrylanguage |
+-----------------+
3 rows in set (0.000 sec)
```

Verificamos los registros de la tabla *countrylanguage*

```
MariaDB [world]> SELECT * FROM countrylanguage;
+-------------+---------------------------+------------+------------+
| CountryCode | Language                  | IsOfficial | Percentage |
+-------------+---------------------------+------------+------------+
| ABW         | Dutch                     | T          |        5.3 |
| ABW         | English                   | F          |        9.5 |
| ABW         | Papiamento                | F          |       76.7 |
| ABW         | Spanish                   | F          |        7.4 |
| AFG         | Balochi                   | F          |        0.9 |
| AFG         | Dari                      | T          |       32.1 |
| AFG         | Pashto                    | T          |       52.4 |
| AFG         | Turkmenian                | F          |        1.9 |
| AFG         | Uzbek                     | F          |        8.8 |
| AGO         | Ambo                      | F          |        2.4 |
| AGO         | Chokwe                    | F          |        4.2 |
| AGO         | Kongo                     | F          |       13.2 |
| AGO         | Luchazi                   | F          |        2.4 |
| AGO         | Luimbe-nganguela          | F          |        5.4 |
| AGO         | Luvale                    | F          |        3.6 |
| AGO         | Mbundu                    | F          |       21.6 |
| AGO         | Nyaneka-nkhumbi           | F          |        5.4 |
| AGO         | Ovimbundu                 | F          |       37.2 |
| AIA         | English                   | T          |        0.0 |
```